

Rotating Away the Location Degeneracy: an Exact Reparameterization for Hamiltonian Monte Carlo in Hierarchical and Joint Models

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Abstract

In a regression model with subject-level random effects, the likelihood identifies the sum of a subject-constant fixed effect and the matching combination of random effects, but not how that sum is split between them. Under the non-centred parameterization that Hamiltonian Monte Carlo (HMC) software uses by default, only the prior pins the split, and the result is a narrow ridge that can slow the sampler on exactly the coefficients an analyst reports, the more so the more each subject’s data inform its own random effects. We characterize the random-effect directions that fixed effects can absorb, for any design, give a construction that does not depend on how the design matrix happens to be parameterized, and state the condition a survival submodel adds in joint longitudinal-survival models. One fixed orthogonal rotation of the standardized random effects, the same for every random-effect column, turns these directions into a small block of sampling coordinates. It is exact for every value of the random-effect covariance: prior, posterior and every reported quantity are unchanged. A dense mass-matrix block over this block and the fixed effects then largely whitens the resulting posterior ridge. In simulation the rotation with its dense block raised the effective sample size per second of the intercept and the time slope by factors of 14 and 21 (geometric means); on two standard joint-model datasets the intercept and subject-constant covariates gained 17–28 times; and the same coefficients gained 5–7 times per gradient evaluation in a linear and a logistic mixed model outside the joint-model setting. A second finding concerns the association parameter of a joint model. For an ideal two-block sampler that draws it exactly from its conditional distribution given the other parameters, the lag-1 autocorrelation equals the fraction of its posterior variance those parameters explain. On real data that fraction is at least 0.85–0.87, mostly shared with the survival parameters (the baseline hazard and the survival coefficients), which caps such a sampler at one effective draw in 13–15 and accounts for about three quarters, on a log scale, of the measured ESS-per-draw gap between HMC and a Metropolis-within-Gibbs implementation.

Keywords: Hamiltonian Monte Carlo; non-centred parameterization; identifiability; preconditioning; mixed models; joint models.

1 Introduction

Mixed models with subject-level random effects are fitted by Hamiltonian Monte Carlo (HMC) and its adaptive variant, the No-U-Turn sampler (NUTS; Hoffman and Gelman, 2014), in general-purpose software such as Stan (Carpenter et al., 2017), brms (Bürkner, 2017), rstanarm and NumPyro (Phan et al., 2019). brms and rstanarm generate the non-centred parameterization of the random effects by default, and it is the standard recommendation for models written directly in Stan or NumPyro: $b = B_{\text{std}}L^\top$ with B_{std} standard normal and L a Cholesky factor of the random-effect covariance. This removes the funnel between the random effects and their scale (Neal, 2003; Betancourt and Girolami, 2015) and is a main reason these samplers work well on hierarchical models.

It leaves a second, simpler problem in place. Suppose the fixed-effects design contains a column that is constant within subject: the intercept, a treatment arm, sex. Adding a constant c to every subject’s random intercept and subtracting c from the fixed intercept leaves every fitted value unchanged. The likelihood identifies the sum; only the priors pin the split, at scale $\sigma_0/\sqrt{N}$ for N subjects when the fixed intercept’s prior is diffuse. The same holds for every subject-constant covariate, whose coefficient trades off against a covariate-weighted combination of the random intercepts, and, whenever there is a random slope on time, for the fixed time slope and the mean random slope. In HMC’s sampling coordinates these directions are spread across all N subjects, the posterior has a ridge along them, and the sampler can mix slowly on the intercept, the subject-constant coefficients and the time slope: the coefficients an analyst reports. In a joint model of `pbc2`, for instance, the posterior correlation between the fixed intercept and the mean random intercept was -0.97 (Section 2.2).

The problem is old and well understood for Gibbs samplers. Hierarchical centring (Gelfand et al., 1995) removes it by sampling the centred effects, at the price of the funnel that non-centring was introduced to avoid, and the choice between the two is the subject of a large literature (Papaspiliopoulos et al., 2003, 2007; Yu and Meng, 2011). A second route removes the directions: sum-to-zero constraints, whose effect on Gibbs and on NUTS Zanella and Roberts (2021) quantify; Stan’s `sum_to_zero_vector`; restricted spatial regression (Reich et al., 2006; Hodges and Reich, 2010); and, exactly, the transformation of Vines et al. (1996), which integrates the grand-mean directions out. The covariate-weighted direction for a random intercept appears in the discussion of Zanella and Roberts (2021) by Yang and Liu (2021) as a device for analysing Gibbs samplers, and Browne et al. (2009) name the problem directly, suggesting as future work that predictors be made orthogonal to the cluster indicators. Section 9.1 places the present work among these.

This paper takes a different route. It keeps the absorbable directions, and makes them sampling coordinates. We construct an orthonormal basis for every direction the fixed effects can absorb, in every random-effect column, and apply one fixed rotation to the standardized random effects so that these directions become a small block U of $k \times q$ coordinates, k being the number of directions and q the number of random-effect columns. A single dense block of the HMC mass matrix over (β, U) then largely whitens the ridge. The rotation acts on the subject index and the covariance factor on

the column index, so they commute: the construction is exact for every value of the random-effect covariance, with no constraint, no correction and no change to any reported quantity.

The motivating application is the Bayesian joint model of a longitudinal marker and a time-to-event outcome (Wulfsohn and Tsiatis, 1997; Henderson et al., 2000; Tsiatis and Davidian, 2004; Rizopoulos, 2012), as implemented by the R package `jmjax`, which samples with NUTS in NumPyro. There the random effects also enter the hazard through the current marker value, and the survival submodel adds a condition on which directions remain absorbable. Joint models also raise a second question. Their primary estimand is the association parameter α , the log hazard ratio per unit of the current marker value. On it, NUTS gave 3 to 8 times the ESS per second of the Metropolis-within-Gibbs sampler of JMbays2 (Rizopoulos et al., 2024), with no reparameterization at all, in a comparison using each package’s own ESS estimator. We explain most of the per-draw part of that gap: for an ideal two-block sampler that draws α exactly given the rest, the lag-1 autocorrelation equals the share of α ’s posterior variance the rest explains (Liu et al., 1994), and in the two joint models we measured that share was large, mostly through the survival parameters, plausibly because α trades off against the level of the log baseline hazard.

Contributions.

1. *The absorbable directions.* An exact characterization of the random-effect directions fixed effects can absorb, for any design and every random-effect column (Proposition 1); a construction that depends only on the column space of the design, with an example where the natural column-by-column check silently misses the slope direction (Example 1); and, for joint models, the condition the survival submodel imposes (Condition S, Proposition 2).
2. *An exact rotation and a small dense metric block.* One Householder rotation of the standardized random effects, costing $O(Nkq)$ per gradient, localizes every absorbable direction in a $(p + kq)$ -dimensional block (Proposition 3, Lemma 1). An analysis of the resulting ridge (Proposition 4) explains why a diagonal metric cannot handle it, why one dense block can, and where a fixed metric is weakest; a stress grid aimed at those weak spots kept the ESS-per-second gains on the intercept and time slope at 5–30 times.
3. *Blocked samplers and the association parameter.* The application of the data-augmentation autocorrelation identity to α (Proposition 5), with a bound that does not need Gaussianity, the measurement of the relevant R^2 on real data (0.85–0.87), and its decomposition, which places most of it in the survival block rather than the random effects.
4. *Evidence.* Simulation grids for one and two random-effect columns, a 100-replicate calibration study, two linear and generalized linear mixed models outside the joint-model setting, and two standard joint-model datasets, including a comparison with JMbays2.

Section 2 sets out the model and the degeneracy, Section 3 the absorbable directions, Section 4 the rotation and the metric, and Section 5 the blocked-sampler result. Sections 6–8 give the

evidence and Section 9 discusses related work, costs and extensions. Proofs not given in the text are in Appendix A; the scripts that reproduce every table and figure are listed in Appendix B.

2 Model and the location degeneracy

2.1 Mixed models and joint models

Index subjects by $i = 1, \dots, N$ and subject i 's longitudinal observations by $j = 1, \dots, n_i$, taken at times t_{ij} . Subject i has a fixed-effects design $X_i \in \mathbb{R}^{n_i \times p}$ and a random-effects design $Z_i \in \mathbb{R}^{n_i \times q}$, and the linear predictor is

$$\eta_{ij} = x_{ij}^\top \beta + z_{ij}^\top b_i, \quad b_i \sim \mathcal{N}_q(0, \Sigma), \quad \Sigma = LL^\top, \quad (1)$$

with $L = \text{diag}(\sigma_b) L_{\text{corr}}$ lower triangular; for a random intercept and slope, $\sigma_b = (\sigma_0, \sigma_1)$ and ρ is their correlation. The response is Gaussian with mean η_{ij} and variance σ_e^2 , or follows a generalized linear model with linear predictor η_{ij} . Collect the random effects as the rows of $B \in \mathbb{R}^{N \times q}$. The non-centred parameterization samples

$$B_{\text{std}} \in \mathbb{R}^{N \times q}, \quad \text{entries iid } \mathcal{N}(0, 1), \quad B = B_{\text{std}} L^\top, \quad (2)$$

and θ collects every other parameter, including β , σ_b and L_{corr} .

In a joint longitudinal-survival model the longitudinal submodel is Gaussian, and subject i 's current underlying marker value

$$m_i(t) = x_i(t)^\top \beta + z_i(t)^\top b_i \quad (3)$$

enters a proportional hazards model,

$$h_i(t) = h_0(t) \exp\{\gamma^\top w_i + \alpha m_i(t)\}, \quad (4)$$

with baseline covariates w_i and association parameter α . Here $x_i(t)$ and $z_i(t)$ are the longitudinal design rows evaluated at time t with subject i 's baseline covariates. In `jmx` the log baseline hazard is a cubic B-spline with a second-order random-walk smoothing prior, and the cumulative hazard is computed by Gauss–Legendre quadrature, so the random effects reach the likelihood through η_{ij} and through $m_i(t)$ at each event time and each quadrature node. The priors on β , γ and α are normal, the random-effect standard deviations have gamma or half-normal priors, and L_{corr} an LKJ prior; none of what follows depends on these choices beyond the prior on B_{std} being iid standard normal.

2.2 The degeneracy

Take a random intercept and slope, $z_{ij} = (1, t_{ij})^\top$, and a fixed-effects design with an intercept, a time column and subject-constant covariates such as age and sex. Three kinds of shift leave every η_{ij} and every $m_i(t)$ unchanged:

- $b_{i0} \mapsto b_{i0} + c$ for all i , with $\beta_0 \mapsto \beta_0 - c$;
- $b_{i0} \mapsto b_{i0} + c \text{ age}_i$ for all i , with $\beta_{\text{age}} \mapsto \beta_{\text{age}} - c$, and likewise for every subject-constant covariate;
- $b_{i1} \mapsto b_{i1} + c$ for all i , with $\beta_{\text{time}} \mapsto \beta_{\text{time}} - c$.

The second kind adds one direction for each subject-constant covariate. The third is present whenever there is a random slope, whatever the covariates. Along each shift the likelihood is exactly invariant, and only the priors locate the split. For the mean intercept direction, with independent random intercepts and a diffuse prior on β_0 , the scale is $\sigma_0/\sqrt{N}$; a proper prior on β_0 narrows it further. We use “degeneracy” for this likelihood invariance. The posterior is proper and the split is identified, weakly: the posterior has a ridge, not a flat direction.

In the coordinates HMC samples, a shift of this kind moves all N entries of one or both columns of B_{std} together (for the mean intercept direction with uncorrelated random effects, each by c/σ_0), against one or a few coordinates of β . With a diagonal metric, the default adaptation in Stan and NumPyro, the kinetic energy rescales coordinates one at a time and cannot represent the covariance along such a direction; a dense metric can, but over all $Nq + p$ coordinates (Section 4). In a fit of the `pbc2` data (Section 8) with an intercept and one subject-constant covariate, the posterior correlation between β_0 and the mean random intercept was -0.971 , and between the covariate’s coefficient and the matching covariate-weighted combination -0.972 . The two affected coefficients had minimum effective sample sizes of 441 and 526 from 4000 draws, against 2893 and more than 4000 for the identified sums. Stan’s user’s guide documents the same mechanism for a model with two intercepts under weak priors (Stan Development Team, 2025).

The degeneracy is not a funnel. A funnel is strongly position-dependent geometry: a scale parameter changes the curvature of the parameters below it, so no single metric fits everywhere. Here, with the variance components held fixed, the posterior of β and the random effects is Gaussian in the linear model (Proposition 4 gives it in closed form for one invariant direction), so the curvature along the ridge is constant. Given the variance components, the problem is linear: the ridge is narrow in a direction a diagonal metric cannot represent, which a fixed Euclidean metric can largely correct once the direction is exposed (Section 4.4 gives what remains when the variance components vary).

3 Absorbable directions

The rotation of Section 4 is exact for any fixed orthogonal matrix. What decides whether it does anything useful is the basis it is built from. The basis should contain every direction the fixed effects

can absorb, so that none is left spread across the N subject coordinates, and only such directions, so that the dense block is not spent on directions the data pin down. This section characterizes them.

Fix a random-effect column c and write $z_i = Z_i e_c \in \mathbb{R}^{n_i}$ for subject i 's entries of it; in this section only, $z_i(t)$ denotes the scalar entry of column c in the design row at time t , rather than the whole row of (3).

Definition 1 (Absorbable direction). A vector $v \in \mathbb{R}^N$ is absorbable for column c if there is a witness $d \in \mathbb{R}^p$ such that the shift $Be_c \mapsto Be_c + v$, $\beta \mapsto \beta + d$ leaves every linear predictor and every marker trajectory, and hence the likelihood, unchanged:

$$X_i d + v_i z_i = 0 \quad \text{for every } i, \quad (\text{L})$$

$$x_i(t)^\top d + v_i z_i(t) = 0 \quad \text{for every } i \text{ and every } t \in [0, \tau]. \quad (\text{T})$$

Write $\mathcal{A}_c$ for the set of absorbable directions and $\mathcal{A}_c^{\text{long}}$ for the set satisfying (L) alone, which is the whole requirement in a mixed model without a survival submodel.

Both sets are linear subspaces of $\mathbb{R}^N$. The likelihood uses $m_i(t)$ only up to each subject's event or censoring time, so requiring (T) on all of $[0, \tau]$ is slightly more than invariance of the likelihood needs; it is what makes the set of directions a property of the model rather than of the observed follow-up. $\mathcal{A}_c$ is therefore a structural set of invariant directions, sufficient for likelihood invariance; it need not be the largest set of shifts that leaves a particular observed likelihood numerically unchanged. The definition concerns the likelihood only: the prior on the random effects is not invariant under these shifts, and that asymmetry is what identifies the split at all.

3.1 Characterization and construction

Let $\mathcal{T} : \mathbb{R}^p \rightarrow \bigoplus_i \mathbb{R}^{n_i}$, $(\mathcal{T}d)_i = X_i d$, map a shift of β to its effect on the stacked linear predictor, and $\mathcal{U}_c : \mathbb{R}^N \rightarrow \bigoplus_i \mathbb{R}^{n_i}$, $(\mathcal{U}_c v)_i = v_i z_i$, do the same for a shift of random-effect column c . Let $P_{\mathcal{T}}$ be the orthogonal projector onto $\text{range } \mathcal{T}$ and, for $z_i \neq 0$, P_{z_i} the projector onto $\text{span}(z_i)$. Let $S_c = \{i : z_i \neq 0\}$ be the subjects with at least one observation at which column c is nonzero, and let $M_c \in \mathbb{R}^{(\sum_i n_i) \times p}$ stack $(I - P_{z_i})X_i$ for $i \in S_c$ and X_i for $i \notin S_c$.

Proposition 1 (Characterization). (a) $\mathcal{A}_c^{\text{long}} = \{v : \mathcal{U}_c v \in \text{range } \mathcal{T}\}$, the preimage of $\text{range } \mathcal{T}$ under $\mathcal{U}_c$, and equals $\ker\{(I - P_{\mathcal{T}})\mathcal{U}_c\}$. It depends on the fixed-effects design only through its column space.

(b) A vector d witnesses some $v \in \mathcal{A}_c^{\text{long}}$ if and only if $d \in \ker M_c$. The witnessed v then has $v_i = -z_i^\top X_i d / z_i^\top z_i$ for $i \in S_c$, and v_i is unrestricted for $i \notin S_c$.

Proof. (a) Condition (L) says $\mathcal{T}d = -\mathcal{U}_c v$ for some d , that is, $\mathcal{U}_c v \in \text{range } \mathcal{T}$, which is equivalent to $(I - P_{\mathcal{T}})\mathcal{U}_c v = 0$. (b) For $i \in S_c$, $X_i d = -v_i z_i$ holds for some scalar v_i if and only if $X_i d \in \text{span}(z_i)$, that is, $(I - P_{z_i})X_i d = 0$, and then $v_i = -z_i^\top X_i d / z_i^\top z_i$. For $i \notin S_c$, (L) reads $X_i d = 0$ whatever v_i is. $\square$

Part (b) is the construction used in practice. In a mixed model without a survival submodel, a subject with $z_i = 0$ has no random effect in column c that enters the likelihood, so we may take S_c to be every subject. Then one singular value decomposition of the $(\sum_i n_i) \times p$ matrix M_c gives a basis of $\ker M_c$, each basis vector d gives one direction $v(d)$, and, if X has full column rank, $d \mapsto v(d)$ is injective, so $\dim \mathcal{A}_c^{\text{long}} = \dim \ker M_c \leq p$. The computation is done once, before sampling. In the implementation every returned direction is also checked against (L) before use, so a loose rank tolerance cannot admit a direction that is not absorbable.

3.2 The survival submodel

In a joint model the requirement (T) quantifies over all $t \in [0, \tau]$, while M_c sees only the observation times.

Condition S (Structural extension). For every $d \in \ker M_c$ and every subject i there is a scalar $v_i(d)$, equal to $-z_i^\top X_i d / z_i^\top z_i$ when $i \in S_c$, such that $x_i(t)^\top d + v_i(d) z_i(t) = 0$ for every $t \in [0, \tau]$. (Under the assumption of Proposition 2, $v_i(d)$ is then unique for every i .)

Proposition 2 (Joint model). *Suppose every subject $i \notin S_c$ has some $t \in [0, \tau]$ with $z_i(t) \neq 0$. Under Condition S, $\mathcal{A}_c = \{v(d) : d \in \ker M_c\}$. For a subject $i \notin S_c$, $v_i(d)$ is determined by (T) at any t with $z_i(t) \neq 0$; in particular it is not free, although the longitudinal data alone leave it free.*

The proof is in Appendix A. Condition S holds for subject-constant, linear-time and covariate-by-time columns: a subject-constant column $(x_i(t)^\top e_j \equiv \text{const})$, absorbed by the intercept column $z_i(t) \equiv 1$, the time column $(x_i(t)^\top e_j = t)$, absorbed by a random slope $z_i(t) = t$ and a covariate-by-time interaction $(c_i t)$. In each the identity holds as a function of t , not by coincidence at the observed times. More generally it holds when, for each subject, the functions $t \mapsto x_i(t)^\top d$ and $t \mapsto z_i(t)$ are known to lie in a finite-dimensional function space determined by its values at the subject's distinct observation times, for example polynomials of degree below n_i (Appendix A). For spline bases this depends on the knots and on where the observations fall, and it can fail for subjects with few observations relative to the flexibility of the basis. It is also cheap to check: the design rows at the event times and quadrature nodes are needed for the likelihood anyway, and `jmjax` evaluates the identity there at every fit and warns if the worst relative residual exceeds 10^{-6} . Proposition 2 also settles how to treat a subject with no observation at which the column is nonzero, such as a baseline-only subject for the slope column: its entry is forced by the survival submodel, not free, and setting it to zero would produce a direction the likelihood does identify.

3.3 A design where the natural check fails

A natural construction, and the one `jmjax` originally used, looks for columns of X that are *individually* subject-constant multiples of z : column j is kept if $X_i e_j = s_i z_i$ for every i , and the basis is the span of the resulting vectors s . This finds a subspace of $\mathcal{A}_c^{\text{long}}$, but not always all of it.

Example 1. Let every subject be observed at the same $m \geq 3$ distinct times, let $Z_i = [1, t]$ with c the slope column, and let $X_i = [1, t + \frac{1}{2}t^2, t^2]$. The ratios of the three columns to t are $1/t$, $1 + t/2$ and t , none constant, so the column-by-column search returns nothing. Yet $d = (0, 1, -\frac{1}{2})$ gives $X_i d = t$, so $v = -\mathbf{1}_N$ satisfies (L) and (T), and $\mathcal{A}_c \neq \{0\}$.

The column space in Example 1 is $\text{span}\{1, t, t^2\}$, exactly what an orthogonal-polynomial or natural-spline mean in time produces; the monomial spelling `time + I(time^2)` is caught, because there the time column appears on its own. So with a spline mean and a random slope, the column-by-column search silently leaves the slope degeneracy in place, while still reporting a basis for the intercept column. By Proposition 1(a) the absorbable set depends only on the column space, and the null-space construction of Proposition 1(b) is invariant to how that space is spanned. It is what `jmjax` now uses; the column-wise search is retained only to report which covariates expose each direction.

Finally, let Q_c be an orthonormal basis of $\mathcal{A}_c$ and $Q \in \mathbb{R}^{N \times k}$ an orthonormal basis of $\text{span} \bigcup_c \mathcal{A}_c$. In the simulation design of Section 6 (an intercept, time, age and sex, with a random intercept and slope) $\mathcal{A}_0 = \text{span}\{\mathbf{1}, \text{age}, \text{sex}\}$ and $\mathcal{A}_1 = \text{span}\{\mathbf{1}\}$, so $k = 3$.

4 The rotation and the metric

4.1 Construction and exactness

Complete Q to an orthogonal matrix $R = [Q \mid Q_\perp] \in \mathbb{R}^{N \times N}$. We use the Householder completion: R is stored as k reflectors, so applying R or $R^\top$ to a vector costs $O(Nk)$ and no $N \times N$ matrix is formed. Instead of B_{std} , sample

$$B_{\text{std}} = R \begin{bmatrix} U \\ V \end{bmatrix} = QU + Q_\perp V, \quad B = B_{\text{std}} L^\top, \quad (5)$$

where $Q \in \mathbb{R}^{N \times k}$, $Q_\perp \in \mathbb{R}^{N \times (N-k)}$, and $U \in \mathbb{R}^{k \times q}$ and $V \in \mathbb{R}^{(N-k) \times q}$ have iid $\mathcal{N}(0, 1)$ entries. Q , and hence R , is computed once from the design matrices before sampling and is held fixed; it does not depend on any sampled parameter.

Proposition 3 (Exactness). *For any fixed orthogonal R : (i) the map $(U, V) \mapsto B_{\text{std}}$ is linear and volume-preserving; (ii) the prior it induces on B_{std} is the original iid $\mathcal{N}(0, 1)$ prior; (iii) the posterior of (B, θ) is identical to that of the non-centred parameterization (2), for every value of L ; and (iv) $U = Q^\top B_{\text{std}}$.*

Proof. $\text{vec}(B_{\text{std}}) = (I_q \otimes R) \text{vec}([U; V])$ and $I_q \otimes R$ is orthogonal, which gives (i), and (ii) because $(I_q \otimes R) \mathcal{N}(0, I_{Nq}) = \mathcal{N}(0, I_{Nq})$. The likelihood depends on (B_{std}, θ) only, and their joint prior is unchanged, so the posterior pushed forward to (B, θ) is unchanged: (iii). (iv) follows from $Q^\top Q_\perp = 0$. $\square$

Because the target distribution is unchanged, the sampler itself is unmodified. NUTS, its step-size and metric adaptation during warm-up, and the argument for its correctness are exactly those of standard NUTS applied to an equivalent parameterization of the same posterior. The rotation changes only which linear combinations of the random effects are sampling coordinates, and therefore what a metric adapted to those coordinates can represent.

Two choices in (5) matter. First, the rotation acts on the sampled, standardized coordinates. Rotating the draws after sampling would change nothing about the sampler’s geometry, and sampling rotated *centred* effects $R^\top B$ would reintroduce the dependence on σ_b that non-centring removes. Second, R acts on the subject index, the same for every column, while L acts on the column index. In vectorized form, $\text{vec}(B) = (L \otimes I_N) \text{vec}(B_{\text{std}})$ and $\text{vec}(B_{\text{std}}) = (I_q \otimes R) \text{vec}([U; V])$, and

$$(L \otimes I_N)(I_q \otimes R) = L \otimes R = (I_q \otimes R)(L \otimes I_N) :$$

the two factors act on different indices and commute. This is why the construction is exact for every value of the covariance with no correction. A rotation of the stacked Nq -vector designed to align particular intercept-slope combinations with coordinate axes would also be exact, but those combinations depend on L , so a fixed rotation of that kind would align them only at one value of the correlation.

For a random intercept alone ($q = 1$), `jmlax` samples the centred effects $b \sim \mathcal{N}(0, \sigma_b^2 I_N)$ directly, and the rotation is applied to b . Because the covariance is a multiple of the identity, $R^\top b$ has the same prior for every σ_b and Proposition 3 holds unchanged.

4.2 Where the degeneracy goes

Lemma 1 (Localization). *Let $v \in \mathcal{A}_c$ with witness d . In the coordinates (β, U, V) , the likelihood-invariant shift $(\beta, Be_c) \mapsto (\beta + d, Be_c + v)$ is*

$$\beta \mapsto \beta + d, \quad U \mapsto U + (Q^\top v) e_c^\top (L^\top)^{-1}, \quad V \mapsto V,$$

for every value of L . Every such shift therefore lies in the coordinate block (β, U) , of dimension $p + kq$.

Proof. Adding v to column c of $B = B_{\text{std}} L^\top$ adds $\Delta = v e_c^\top (L^\top)^{-1}$ to B_{std} . Every column of Δ is a multiple of $v \in \text{span}(Q)$, so $Q_\perp^\top \Delta = 0$ and $Q^\top \Delta = (Q^\top v) e_c^\top (L^\top)^{-1}$. $\square$

Without the rotation, the same shift has a component on every one of the N rows of B_{std} . With it, the likelihood-invariant directions mix the intercept and slope columns of U in proportions set by the sampled L , but always stay inside the fixed block (β, U) . This is why every column is rotated by the same Q : rotating the intercept column alone captures the directions only when the random effects are uncorrelated. Because the same Q , a basis of the union of the $\mathcal{A}_c$, is used for every column, U always has kq coordinates, and a few of them may not be invariant directions for

their own column (in Section 6’s design, $k = 3$ and 4 of the 6 coordinates of U are). Including them in the dense block costs little and keeps the construction independent of L .

4.3 The ridge, and the dense block

Invariance of the likelihood is not degeneracy of the posterior: along each likelihood-invariant direction the priors on β and B_{std} still locate the split, and the posterior is proper but weakly identified along it. The next result describes the ridge this leaves in one plane.

Proposition 4 (The ridge). *Consider one invariant direction in the reduced model that holds every other coordinate fixed. Let $\tilde{\beta}$ be the coordinate of β along the witness, with a flat prior, and u the matching coordinate of U , with prior $\mathcal{N}(0, 1)$, and suppose the likelihood informs only $s = \tilde{\beta} + \kappa u$, with precision τ . Then*

$$(i) \text{ corr}(\tilde{\beta}, u \mid y) = -\{1 + 1/(\tau\kappa^2)\}^{-1/2};$$

(ii) *under the ideal diagonal metric, equal to the exact marginal posterior variances (the quantity warm-up adaptation estimates), the whitened covariance of the plane has eigenvalues $1 \pm |r|$, r the correlation in (i). Its wide direction has variance at most 2, but its narrow direction has variance $1 - |r| \approx 1/(2\tau\kappa^2)$;*

(iii) *a dense metric over $(\tilde{\beta}, u)$ whitens the plane exactly;*

(iv) *for the mean direction of the random intercepts, $\kappa = \sigma_0/\sqrt{N}$ and $\tau = \sum_i n_i/\sigma_e^2$, so $\tau\kappa^2 = \bar{n}\sigma_0^2/\sigma_e^2$, the information each subject carries about its own level. It does not depend on N .*

Proof. The posterior precision of $(\tilde{\beta}, u)$ is $\tau(1, \kappa)^\top(1, \kappa) + \text{diag}(0, 1)$, whose inverse is

$$\tau^{-1} \begin{pmatrix} 1 + \tau\kappa^2 & -\tau\kappa \\ -\tau\kappa & \tau \end{pmatrix};$$

this gives (i). Whitening a 2×2 covariance by its diagonal gives its correlation matrix, whose eigenvalues are $1 \pm |r|$: (ii). (iii) is the definition of a dense metric. For (iv), moving the mean direction by u moves every subject’s level by $\sigma_0 u/\sqrt{N}$, and the data inform a common shift of all levels with precision $\sum_i n_i/\sigma_e^2$. $\square$

Heuristically, for a Gaussian target the narrow direction bounds the stable leapfrog step size and the wide one sets the trajectory length, so the number of leapfrog steps per independent draw grows like the square root of their ratio, $\{(1 + |r|)/(1 - |r|)\}^{1/2} \approx 2\sqrt{\tau\kappa^2}$ (Neal, 2011). Proposition 4 itself is exact for the mean intercept direction, whose witness is the single coordinate β_0 . For other directions the witness mixes several coordinates of β , and (ii)–(iii) then describe the whole (β, U) block. The rotation alone therefore trades a direction spread over N coordinates for one concentrated in a few, but under a diagonal metric that direction is still narrow, and narrowest where the degeneracy is worst: many visits per subject, or little noise. A dense metric block

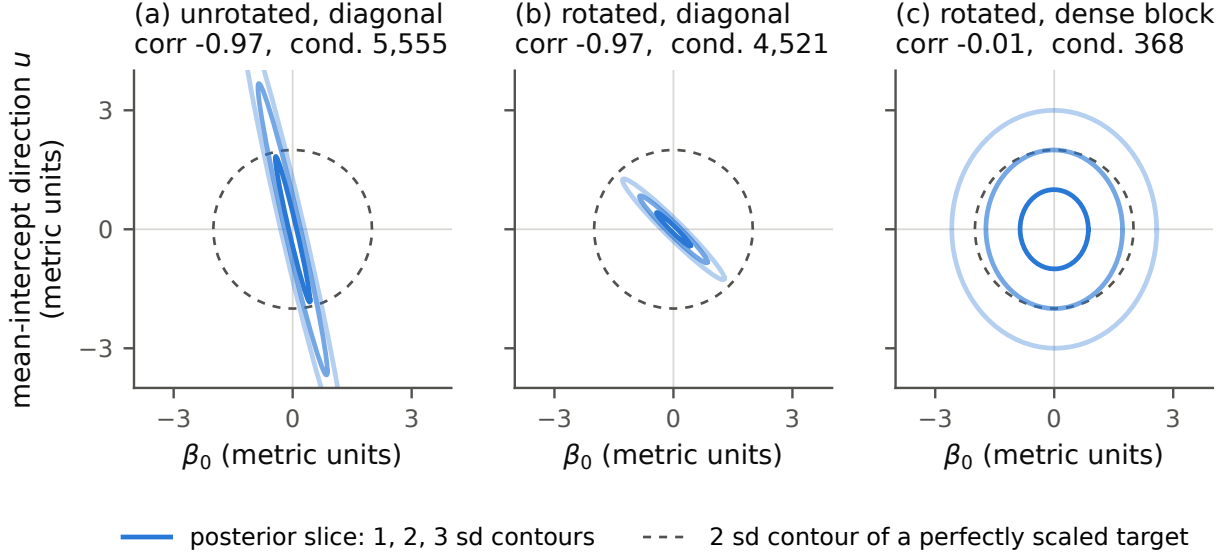


Figure 1: The location ridge and what removes it. Exact conditional posterior of the fixed intercept β_0 and the mean intercept direction u , through the mode, in the units the HMC metric sees them (Gaussian longitudinal submodel, variance components fixed; $N = 300$, 2–11 visits, $\rho = 0.3$, $\sigma_e = 0.3$). Solid: 1, 2 and 3 standard deviation contours; dashed: the 2 standard deviation contour of a perfectly scaled target. (a) Unrotated coordinates, diagonal metric: u is spread over all N random intercepts and trades off almost one for one with β_0 . (b) The rotation makes u a single coordinate but leaves the ridge. (c) The dense metric block over (β, U) whitens it. “cond.” is the condition number of the whole whitened posterior.

over (β, U) whitens it. Without the rotation, the same fix would need a dense metric over all Nq random-effect coordinates: an $Nq \times Nq$ matrix to estimate during warm-up, $O(N^2q^2)$ storage and work for each metric-vector product, and more to factorize it. Section 7 measures this alternative directly.

Figure 1 shows this in the exact Gaussian posterior of the longitudinal submodel with the variance components fixed at the simulation values of Section 6 ($N = 300$, 2–11 visits, $\rho = 0.3$, $\sigma_e = 0.3$). The conditional correlation of β_0 and the mean intercept direction is -0.97 in the unrotated coordinates and still -0.97 after rotation with a diagonal metric; the dense block brings it to -0.01 . The condition number of the whole whitened posterior, the ratio of the largest to the smallest eigenvalue of $M^{-1/2}\Sigma M^{-1/2}$ with Σ the exact posterior covariance and M the metric (diag Σ , or diag Σ with its (β, U) block replaced by that of Σ), falls from 4,521 to 368. Across the designs examined, the dense block cut it by factors of 5 to 70, for example from 12,037 to 174 with 8–20 visits per subject; what remains is conditioning any fit has, such as the correlation between each subject’s own intercept and slope.

4.4 What a fixed metric cannot do

By Lemma 1, the ridge’s orientation depends on the sampled L : on σ_0 through $\kappa \propto \sigma_0$, and, for directions shared between the intercept and slope columns, on the correlation ρ . NUTS adapts one fixed metric. For the σ_0 part the loss can be estimated. Let δ be the posterior coefficient of variation of σ_0 ; in an idealized variance-component model with N groups it is roughly $1/\sqrt{2N}$, and we use that value only to set the scale. A relative change δ in $\kappa \propto \sigma_0$ tilts the plane’s narrow direction, in the units of a metric fitted at the original value, by an angle of order $\delta\sqrt{\tau\kappa^2}$, so a metric that whitens the plane exactly at one value of σ_0 leaves a condition number of $1 + O(\delta\sqrt{\tau\kappa^2})$ elsewhere in the posterior. With $\delta \approx (2N)^{-1/2}$ and $\tau\kappa^2 = \bar{n}\sigma_0^2/\sigma_e^2$ this is

$$1 + O\left(\sqrt{\bar{n}\sigma_0^2/(\sigma_e^2N)}\right),$$

which is small when there are many subjects relative to the information each carries, the usual regime for longitudinal data. In the exact Gaussian model of Figure 1 the worst block condition number over $\sigma_0(1 \pm \delta)$ was 1.76, 1.50 and 1.31 for $N = 150, 300$ and 600 with 2–11 visits, and 2.71, 2.05 and 1.64 with 8–20 visits. The analysis does not cover the ρ part, which grows as $|\rho| \rightarrow 1$. These are the two regimes Section 6.4 stress-tests. For $q = 1$ in centred coordinates, the ridge’s orientation does not involve σ_b ; only the scale of the rotated coordinate’s prior does, a mismatch of order δ rather than $\delta\sqrt{\tau\kappa^2}$.

4.5 Implementation and cost

`jmxax` implements (5) in NumPyro: U and V are the sampled sites, B_{std} is a deterministic function of them computed with q applications of the k reflectors, and the mass matrix is block-diagonal: dense on (β, U) and diagonal elsewhere, specified through NumPyro’s option for dense blocks over groups of sites and adapted during warm-up in the usual way. The extra cost per gradient is $O(Nkq)$. In Section 6’s design the dense block has $p + kq = 10$ coordinates, and wall time per fit was 0.83–1.10 times that of the unrotated fit. The basis is computed once from the design matrices, before sampling. The rotation and the dense block are the package’s default for NUTS fits with a random intercept only, or with $q \geq 2$ correlated random effects.

5 Blocked samplers and the association parameter

The location degeneracy does not involve α . Yet on α , `jmxax`’s NUTS was 3–8 times more efficient per second than JMBayes2 even without the rotation: 5.7–8.1 times across a replicated simulation sweep with 1 to 12 covariates, and 3.2 and 7.6 times on the two datasets of Section 8. JMBayes2 updates the parameters in blocks, with adaptive random-walk Metropolis steps for the coefficients. This section argues that, on the two datasets measured, about three quarters (on a log scale) of the ESS-per-draw gap on α is what even an ideal two-block sampler would incur, so that most of it is a property of blocked updating rather than of either implementation.

Write ψ for every parameter other than α : the random effects, β , γ , the baseline-hazard coefficients and the variance components.

Proposition 5 (Blocked updates of α). *Consider the two-block sampler that draws α exactly from $p(\alpha \mid \psi, y)$ and then ψ exactly from $p(\psi \mid \alpha, y)$, and let*

$$r = \frac{\text{Var}\{E(\alpha \mid \psi, y)\}}{\text{Var}(\alpha \mid y)} = 1 - \frac{E\{\text{Var}(\alpha \mid \psi, y)\}}{\text{Var}(\alpha \mid y)},$$

the fraction of α 's posterior variance explained by the other parameters, and $R^2 \leq r$ the squared multiple correlation of α on ψ .

- (i) *At stationarity the lag-1 autocorrelation of the α chain is r .*
- (ii) *Its integrated autocorrelation time is at least $(1+r)/(1-r)$, so its effective sample size per draw is at most $(1-r)/(1+r) \leq (1-R^2)/(1+R^2)$.*
- (iii) *If the posterior is Gaussian, $r = R^2$, the α chain is $AR(1)$ with coefficient R^2 , and the bound in (ii) is attained.*

Proof. The α chain of a two-block sampler is a data-augmentation chain (Liu et al., 1994). Let P map a function h of α to the function $(Ph)(\psi) = E\{h(\alpha) \mid \psi, y\}$, and P^* map g to $(P^*g)(\alpha) = E\{g(\psi) \mid \alpha, y\}$; these are adjoint in the L^2 inner products of the posterior. The transition operator of the α chain is $K = P^*P$, so K is self-adjoint and $\langle Kh, h \rangle = \|Ph\|^2 \geq 0$. With $h(\alpha) = \alpha - E(\alpha \mid y)$, the lag-1 autocovariance is $\langle Kh, h \rangle = \text{Var}\{E(\alpha \mid \psi, y)\}$, which gives (i); the second expression for r is the law of total variance, and $R^2 \leq r$ because the best linear predictor of α explains no more variance than $E(\alpha \mid \psi, y)$. For (ii), the spectral theorem writes the lag- k autocorrelation of α as $\rho_k = \int \lambda^k \mu(d\lambda)$ for a probability measure μ on $[0, 1]$, so Jensen's inequality gives $\rho_k \geq \rho_1^k = r^k$ and $1 + 2 \sum_{k \geq 1} \rho_k \geq (1+r)/(1-r)$; the function $(1-x)/(1+x)$ is decreasing. For (iii), $E(\alpha \mid \psi)$ is linear in ψ and $E(\psi \mid \alpha)$ linear in α ; composing the two regressions gives $\alpha_{t+1} = c + R^2 \alpha_t + \varepsilon_t$ with ε_t independent of the past, since the coefficient is $\text{Cov}(\alpha, \psi) \Sigma_\psi^{-1} \text{Cov}(\psi, \alpha) / \text{Var}(\alpha) = R^2$, and then $\rho_k = (R^2)^k$. $\square$

Part (ii) is what makes a measured linear R^2 usable: it bounds the efficiency of any exact two-block sampler from above, whatever the shape of the posterior. The proposition is an application of known theory; the two-block sampler is a benchmark, not a model of JMBayes2. Finer blocking and random-walk Metropolis steps can be expected to make a real implementation slower still (Liu et al., 1994; Roberts and Sahu, 1997), though neither is guaranteed to in every case. The contrast with HMC, which moves α and ψ together, is a heuristic rather than a theorem. Under a fixed metric, a correlation between them is part of the posterior's shape and is paid for in leapfrog steps, which for a Gaussian target grow with the square root of the condition number (Neal, 2011). A single correlation at level R^2 contributes a condition number $(1 + \sqrt{R^2})/(1 - \sqrt{R^2})$, so the cost grows at most like $1/\sqrt{1 - R^2}$, against $1/(1 - R^2)$ for the blocked sampler. Often it grows not at

all, because in a joint model the step size is already set by some other narrow direction, such as the spline coefficients or the variance components. In a Gaussian check, comparing a simulated two-block Gibbs sampler with a stylised model of HMC (exact Hamiltonian dynamics, random integration time), the two-block sampler’s lag-1 autocorrelation matched R^2 to within 0.004, and its integrated autocorrelation time went 3.0, 9.0, 27.6, 39.0 and 199 as R^2 went 0.5, 0.8, 0.93, 0.95 and 0.99. The HMC model’s stayed between 1.6 and 2.1, at a cost of 4.8 to 46.8 gradient evaluations per independent draw; with an unrelated narrow ridge already present, the step count was 22.2 at every R^2 from 0 to 0.99.

5.1 How large R^2 is in a joint model

Two mechanisms can make R^2 large. The first is errors in variables: given the random effects, each marker trajectory $m_i(t)$ is known and α is the coefficient of a known time-varying covariate, which the data pin sharply; marginally, α also carries the uncertainty in every trajectory. The second is the level of the hazard. Shifting $m_i(t)$ by a constant for every subject changes $\alpha m_i(t)$ by an amount the log baseline hazard absorbs, so α is correlated with the baseline-hazard level unless the marker is centred.

We estimated the linear explained variance R^2 by an ordinary least-squares regression of the 20,000 draws of α from a long rotated fit (4 chains of 5000 draws, close to independent) on every other sampled quantity: the random effects, β , γ , the baseline-hazard coefficients and the variance components, 960 regressors for **aids** and 649 for **pbc2** after dropping constant columns. We report the adjusted R^2 ; with 20,000 draws the adjustment changes it by less than 0.01. We also regressed α on each block alone. On **aids**, $R^2 = 0.873$; the survival block alone (γ and the baseline-hazard coefficients) explained 0.86 of α ’s posterior variance and the random effects alone 0.30. On **pbc2** the figures were 0.853, 0.82 and 0.20. In these two datasets, most of the linear explained variance is attributable to the survival block rather than the random effects. By Proposition 5(ii), the linear R^2 is a lower bound on the fraction r , so the efficiency it implies for an ideal blocked sampler is an upper bound.

Figure 2 puts the measurements against the prediction. At the measured R^2 an ideal two-block sampler keeps at most 0.068 (**aids**) and 0.079 (**pbc2**) effective draws per draw, one in 13–15. JMBayes2 achieved 0.035 and 0.028, and jmjax’s NUTS 1.05 and 1.72 in the same long fit. On a log scale, the ideal benchmark accounts for 80% of the measured ESS-per-draw gap on **aids** (15.5 of 30 times) and 75% on **pbc2** (21.7 of 61 times); the rest, a factor of two to three, is attributable to the implementation. A blocked sampler could escape most of the structural part by updating α jointly with γ and the baseline-hazard coefficients: with only the random effects conditioned away, the ideal ESS per draw would be about 0.54 and 0.67. But that block has 10–20 strongly correlated dimensions without exact conditional draws, and an adaptive random-walk proposal on it is itself slow. HMC moves α and ψ along a joint trajectory without an explicitly designed block, although how well it does so still depends on the metric and on the rest of the geometry.

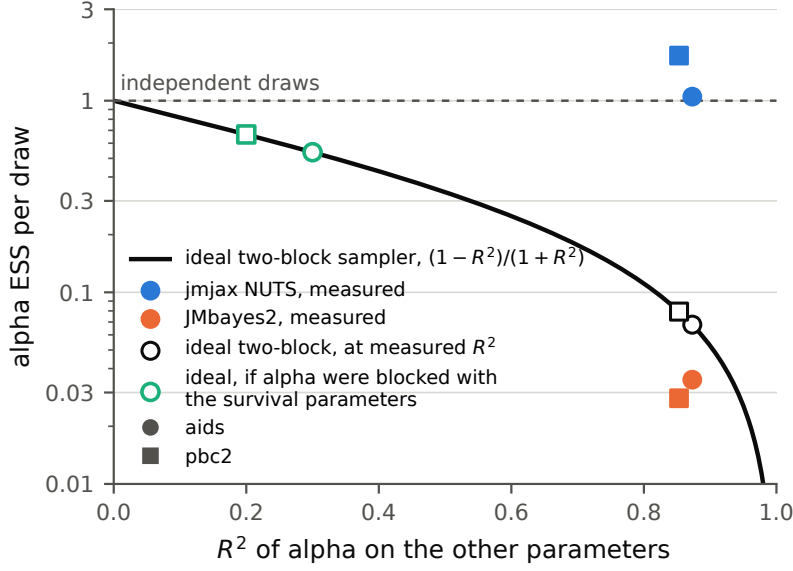


Figure 2: Effective sample size per draw for α . Curve: an ideal two-block sampler, $(1 - R^2)/(1 + R^2)$ (Proposition 5). Black open markers: the ideal sampler at the measured R^2 (0.873 for **aids**, 0.853 for **pbc2**). JMbays2 does two to three times worse than that ideal; the NUTS draws are effectively independent. Green open markers: the ideal if α were updated in one block with the survival parameters, so that only the random effects are conditioned away ($R^2 = 0.30$ and 0.20).

6 Simulation

6.1 Design

Data were simulated from a joint model with a random intercept and slope, longitudinal mean $\beta_0 + \beta_1 t + \beta_{\text{age}} \text{age}_i + \beta_{\text{sex}} \text{sex}_i$ with $\beta_0 = 2$ and $\beta_1 = 0.5$, $\text{age}_i \sim \mathcal{N}(0, 10^2)$ and $\text{sex}_i \sim \text{Bernoulli}(0.5)$, $\sigma_b = (0.8, 0.3)$, $\sigma_e = 0.3$, and a Weibull hazard (shape 1.3) with $\alpha = 0.5$ and a treatment effect $\gamma = -0.3$, calibrated to an event rate of about one half. Visits were scheduled at regular intervals up to $t = 10$ and stopped at the event or censoring time. All fits used the spline baseline hazard of Section 2.1, so the fitted model does not contain the true hazard exactly. Each design was simulated with three seeds, and every seed was fitted twice on the same data: unrotated, and with the rotation and its dense block. Both fits report the same parameters, β included, so the comparison is on the quantities a user reports. The design is paired: both arms see the same data. Effective sample size (ESS) is NumPyro’s (version 0.15) multi-chain autocorrelation estimator, truncated by Geyer’s initial monotone sequence and without rank normalization (Geyer, 1992; Gelman et al., 2013), and $\hat{R}$ is its split-chain version. ESS per second divides by the wall time of the whole MCMC run, warm-up and compilation included; ESS per gradient divides by the number of leapfrog steps in the sampling phase. Ratios are rotated over unrotated, paired by seed, with the geometric mean over seeds reported for each design and the per-seed range in the text.

The core grid crossed about 9 against about 17 visits per subject with $\rho \in \{0, 0.3, 0.6, 0.9\}$ at

Table 1: Core grid: ESS per second for the intercept and the time slope (means over 3 seeds), and the rotated/unrotated ratio (geometric mean of the paired per-seed ratios).

visits	ρ	N	intercept			time slope			α
			unrot.	rotated	ratio	unrot.	rotated	ratio	ratio
9	0.0	300	25.9	354.4	14.2	9.1	304.9	36.4	0.85
17	0.0	300	12.1	229.4	19.3	4.0	234.9	59.2	1.32
9	0.3	300	30.0	253.3	8.6	19.0	251.1	13.4	0.72
17	0.3	300	11.3	248.5	22.2	6.4	262.5	41.9	1.33
9	0.6	300	26.1	245.7	9.4	29.9	229.3	7.7	1.00
17	0.6	300	9.6	231.6	24.0	10.0	251.2	24.9	1.59
9	0.9	300	20.4	220.7	11.1	20.3	250.3	12.8	0.90
17	0.9	300	8.9	218.9	24.9	8.6	217.7	25.6	1.59
9	0.3	150	41.8	323.7	7.6	32.6	420.4	13.1	0.76
9	0.3	600	15.9	132.0	8.2	8.8	151.7	17.7	0.77

$N = 300$, and added $N \in \{150, 600\}$ at 9 visits and $\rho = 0.3$: ten designs, 2 chains of 500 warm-up and 1000 sampling iterations.

6.2 Efficiency

Table 1 gives the results. Over all 30 paired fits the geometric-mean gain was 13.5 for the intercept (per seed 4.5–31.6) and 21.1 for the time slope (5.1–70.8). Wall time was essentially unchanged (ratio 1.03, range 0.83–1.10), and the rotated fit took slightly fewer leapfrog steps per iteration (0.89 times; 0.49–1.02), so per gradient evaluation the gains were 15.6 for the intercept and 24.3 for the time slope. Almost all of the gain is in ESS per draw (13.9 and 21.7 times): under the rotation the median ESS over 2000 draws was about 3,450 for both coefficients, more than the number of draws, against 180–235 unrotated, that is, about 1.7 against 0.1 ESS per draw. There were no divergences, and the worst $\hat{R}$ on a reported coefficient was 1.002 rotated and 1.044 unrotated. Posterior means agreed to a median of 0.027 posterior standard deviations, and the largest paired difference was 2.7 Monte Carlo standard errors over 90 pairs. Posterior standard deviations agreed too: rotated over unrotated, 1.01 for the intercept (per seed 0.94–1.10) and 0.98 for the time slope (0.91–1.06), within the Monte Carlo error of the unrotated fit’s short effective samples.

The pattern follows Proposition 4. The gain is largest with dense visits, where each subject carries the most information about its own random effects (19–25 times for the intercept, 25–59 for the slope), and does not grow with N (7.6 at $N = 150$, 8.2 at $N = 600$), because the ridge is set by per-subject information, not by the number of subjects. Figure 3 shows all three grids against that information, $\bar{n} \sigma_0^2 / \sigma_e^2$.

The association parameter, which the degeneracy does not involve, is not untouched: its ratio split by design, 0.72–1.00 in the six designs with about 9 visits and 1.32–1.59 in the four with about 17 (geometric mean 1.04). We do not have an established explanation. On real data (Section 8) the effect on α tracked whether the degeneracy had been setting NUTS’s trajectory length, which is a plausible reading here too but was not measured.

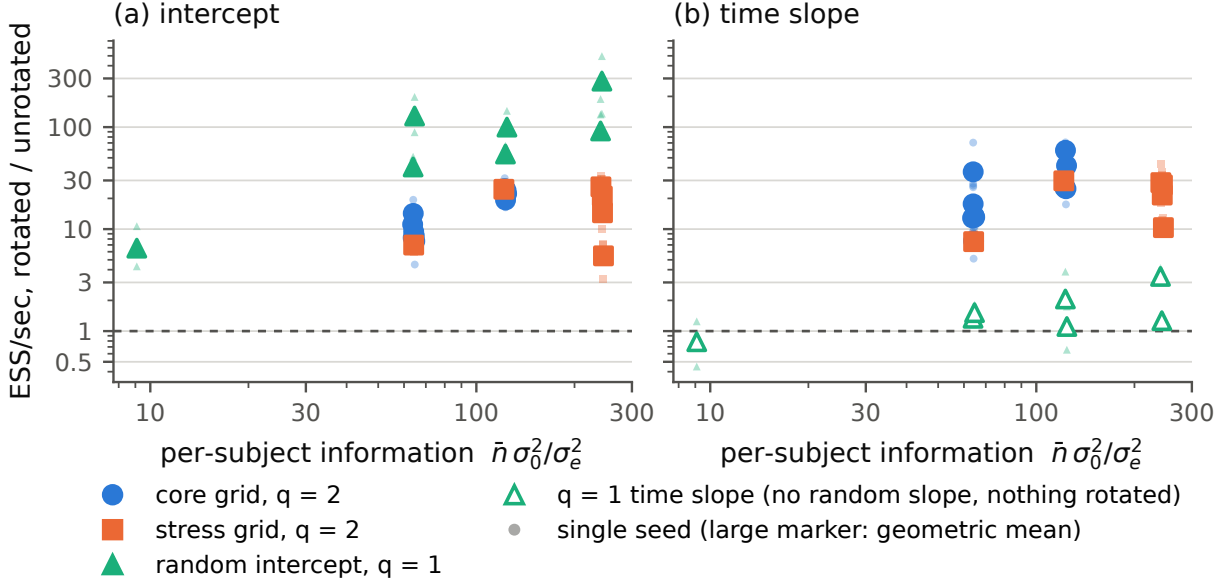


Figure 3: ESS per second of the reported coefficient, rotated over unrotated fit on the same simulated data, against the per-subject information $\bar{n} \sigma_0^2 / \sigma_e^2$ ($\sigma_0 = 0.8$). Large markers: geometric mean over three seeds in one design; small: single seeds. Core and stress grids have a random intercept and slope ($q = 2$); the $q = 1$ grid has a random intercept only. In (b), open triangles are the $q = 1$ time slope, which has no random effect and so nothing to rotate. The $q = 1$ intercept ratios are at the standard run length, where most unrotated fits had not converged; Section 6.3 gives a comparison between converged fits.

6.3 Random intercept only

The same simulator with $\sigma_{b1} = 0$, fitted with a random intercept only, gave seven designs: about 9, 17 and 35 visits, $N \in \{100, 300\}$, and one noisier design with $\sigma_e = 0.8$. The intercept gained 6.5–282 times and the subject-constant coefficient of age 6.6–207 times; the time slope, which has no absorbable direction when there is no random slope, moved only through the step size (0.79–3.4 times). At this run length, however, 17 of the 21 unrotated fits had some parameter with $\hat{R}$ above 1.05, the intercept’s reaching 1.27, while every rotated fit had $\hat{R} \leq 1.004$, so these ratios compare a converged fit with unconverged ones. Posterior means agreed to a median of 0.025 posterior standard deviations (maximum 0.175).

To obtain a comparison between converged fits, we reran three designs ($N = 300$ with about 9, 17 and 35 visits) and extended the unrotated fit to 2, 4, 8, 16 and 32 times the standard length, with two seeds taken to the longest runs. A convergence criterion was fixed in advance: split $\hat{R} \leq 1.01$ on every monitored parameter and $\text{ESS} \geq 400$ for the intercept, age, the time slope and α . Every rotated fit met it at the standard length. The unrotated fit met it first at 16 times the standard length with 9 visits, at 16 and 32 times with 17 visits, and not by 32 times with 35 visits, where its intercept had an ESS of 304–377 from 64,000 draws with $\hat{R} = 1.002$. Its ESS per draw for the intercept did not change with run length: 0.009–0.023, 0.007–0.020 and 0.002–0.008 in the

Table 2: Stress grid: ESS-per-second ratios, rotated over unrotated (geometric means over 3 seeds).

visits	N	ρ	intercept	time slope	α
35	50	0.30	5.5	10.3	1.60
35	50	0.95	21.1	21.6	2.42
35	100	0.30	14.5	27.1	1.58
35	100	0.95	26.0	28.6	1.73
9	100	0.95	7.0	7.5	0.74
17	300	0.95	24.6	29.7	1.09

three designs over runs from 1 to 32 times the standard length, against 2.1–2.5, 0.9–1.9 and 1.1–2.4 for the rotated fit. The per-draw advantage, of order 100 times, is therefore a property of the stationary chains, not of short runs. In wall time, the unrotated fit needed 3.5 and 6.5–6.6 times as long as the rotated fit to meet the criterion in the first two designs, and more than 10–14 times in the third. These ratios are much smaller than the per-draw ones because the rotated fit’s time at the standard length is mostly warm-up and compilation, and it exceeded the ESS target by a factor of 5 to 12. Between the first fits of each arm to meet the criterion, posterior means agreed to a median of 0.02 posterior standard deviations (maximum 0.11) and standard deviations to within 6%.

6.4 Stress grid

Section 4.4 names two regimes where the rotation should be weakest: few subjects each carrying a lot of information, and strongly correlated random effects. A stress grid targeted both, with about 35 visits crossed with $N \in \{50, 100\}$ and $\rho \in \{0.3, 0.95\}$, plus $\rho = 0.95$ at 9 visits ($N = 100$) and at 17 visits ($N = 300$). The criterion fixed before the run was that the rotation must not fall below the unrotated fit on the intercept or time slope in any design. It did not (Table 2); the lowest design was 5.5 times and the lowest single seed 3.2 times. Per gradient evaluation the gains were 16.7 for the intercept and 22.0 for the time slope (geometric means over all designs), and the rotated fit took 0.81 times as many leapfrog steps per iteration.

The design with the fewest subjects and the most information per subject gained least, as the fixed-metric residual predicts. Strong correlation did not reduce the rotation’s advantage: at $\rho = 0.95$ the gains were as large as or larger than at $\rho = 0.3$, partly because the unrotated fit degrades faster. Nor did it hurt the rotated fit in absolute terms: with 35 visits its ESS per draw for the intercept was 1.25 at $\rho = 0.95$ against 0.68 at $\rho = 0.3$ with $N = 50$, and 1.50 against 1.18 with $N = 100$. The predicted weakness is visible only against a variant that removes the absorbable directions from the likelihood’s coordinates and restores the reported coefficients with an exact correction (Section 9.1). That variant has no fixed-metric residual, and in the 35-visit designs it was 1.4–2 times more efficient than the rotation, while being about equal to it with 9 visits. Posterior means agreed with the unrotated fit’s to a median of 0.029 posterior standard deviations (maximum 0.166, 90 pairs), there were no divergences, and wall time was 0.86–1.08 times the unrotated fit’s (geometric means per design). α , which the degeneracy does not involve,

Table 3: Calibration, $N = 300$, 100 replicates: coverage of 95% intervals, ratio of mean posterior SD to the SD of posterior means, and bias in Monte Carlo standard errors.

estimand	unrotated			rotated		
	cover	SD ratio	bias/se	cover	SD ratio	bias/se
β_0	0.99	1.164	1.68	0.99	1.150	1.55
β_{time}	0.97	1.027	-0.30	0.96	1.038	-0.32
α	0.93	0.922	2.97	0.93	0.917	2.97
σ_{b0}	0.94	0.914	0.63	0.95	0.906	0.68

fell to 0.74 in the 9-visit design (Table 2).

6.5 Calibration

Proposition 3 says the rotation cannot change the posterior, so a calibration study checks the implementation rather than the method. With $N = 300$ and 100 replicates, every one of which passed an $\hat{R} \leq 1.05$ check on the regression parameters (so none was excluded), coverage of 95% intervals agreed within 0.01 and the ratio of mean posterior standard deviation to the standard deviation of posterior means within 0.015 for every estimand (Table 3); the Monte Carlo standard error of a 95% coverage is about 0.011. Two features appear in both arms and belong to the model and design: slightly wide intervals for β_0 , and an upward bias of about 3 Monte Carlo standard errors (2.8%) in α . The latter is consistent with ordinary finite-sample bias. In a separate study (`dev/study_alpha.bias.R`, Appendix B), neither the spline baseline hazard nor the simulator’s hazard calibration explained it (paired changes of -0.0011 and -0.0016 , standard error about 0.0025), and at $N = 1200$ it fell to 1.1%, 1.7 Monte Carlo standard errors, with coverage 0.92.

7 Beyond joint models

Nothing in Sections 3–4 depends on the survival submodel except Condition S. To check that the gains are not specific to joint models, we applied the construction, in a standalone NumPyro program that does not use `jmxax`, to two classical mixed models, each with a subject-constant covariate:

- **Orthodont** (Potthoff and Roy, 1964; Pinheiro and Bates, 2000): a Gaussian linear mixed model for the distance measure of 27 children at 4 ages, $\text{distance} \sim \text{age} \times \text{male} + (1 + \text{age} \mid \text{child})$ with age centred. Sex is subject-constant, so both random-effect columns have absorbable directions $\{\mathbf{1}, \text{male}\}$, $k = 2$.
- The toenail onychomycosis trial (De Backer et al., 1998; Lesaffre and Spiessens, 2001): a logistic mixed model for 1908 binary outcomes on 294 patients, $y \sim \text{trt} \times \text{time} + (1 \mid \text{patient})$. Treatment is subject-constant, so the absorbable directions are $\{\mathbf{1}, \text{trt}\}$.

Table 4: Two mixed models: ESS per 1000 gradient evaluations relative to the unrotated fit with a diagonal metric, and in brackets ESS per second (geometric means over 3 seeds; ranges over the coefficients in a group). The last three columns give leapfrog steps per draw, seconds per run (whole run) and divergences over the 3 seeds.

	coefficients					
arm	intercept, subject-const.	absorbed by slope	variance comp.	steps	sec	div.
Orthodont (LMM, $q = 2$, $N = 27$; subject-constant: male; slope-absorbed: age, age $\times$ male)						
unrotated	1	1	1	30.7	1.9	0
rotated, diagonal	2.3–2.4 (2.6–2.7)	0.9 (1.1)	0.9 (1.0–1.1)	30.9	1.6	0
rotated, dense block	7.1 (4.7)	2.4–2.5 (1.6)	1.3–1.6 (0.8–1.0)	16.5	1.5	0
unrotated, full dense	5.5–5.6 (2.0–2.1)	1.7–1.8 (0.6)	1.2–2.0 (0.5–0.7)	18.9	3.0	2
centred	0.4 (0.6–0.7)	0.4–0.5 (0.7)	0.1–1.7 (0.1–2.6)	52.5	1.7	34
toenail (logistic GLMM, $q = 1$, $N = 294$; subject-constant: trt; not absorbable: time, trt $\times$ time)						
unrotated	1	—	1	31.0	7.7	0
rotated, diagonal	2.5–2.8 (2.3–2.5)	—	0.9 (0.9)	31.0	8.5	0
rotated, dense block	5.5–6.7 (4.2–5.1)	—	1.1 (0.9)	21.2	6.8	0
unrotated, full dense	3.2–3.7 (0.2–0.3)	—	0.9 (0.07)	47.6	166.8	0
centred	0.8–1.1 (0.5–0.8)	—	0.4 (0.2)	30.6	10.8	0

Both use the non-centred parameterization with $\mathcal{N}(0, 10^2)$ priors on β , half-normal(5) priors on the standard deviations and an LKJ(2) prior on the correlation, which is the parameterization brms and rstanarm use by default. Five arms were run on each, with 4 chains of 1000 warm-up and 1000 sampling iterations and 3 seeds, all in the same program on the same data:

- *unrotated*, diagonal metric (the reference);
- *rotated*, *diagonal*: the rotation alone;
- *rotated*, *dense block*: the method, a dense metric on (β, U) only;
- *unrotated*, *full dense*: no rotation, and a dense metric over every sampled coordinate, the obvious alternative way to let the sampler learn the ridge;
- *centred*: the random effects themselves are sampled, $b_i \sim \mathcal{N}(0, LL^\top)$, with a diagonal metric, the classical remedy for the location degeneracy (Gelfand et al., 1995).

Because each run takes seconds, we report ESS per 1000 gradient evaluations, which does not depend on the machine, and ESS per second of the whole run (warm-up and compilation included) as the practical measure.

Table 4 gives the results. With the method, the intercept and the subject-constant coefficients gained 5.5–7.1 times per gradient and 4.2–5.1 times per second, and the coefficients absorbed by the random slope in **Orthodont** 2.4–2.5 times per gradient. The rotation with a diagonal metric alone supplied a factor of 2.3–2.8 on the intercept and subject-constant coefficients and nothing on the slope-absorbed ones; the dense block supplied the rest, as Proposition 4 predicts. The coefficients with no absorbable direction (toenail’s time and treatment-by-time) moved by 1.3–1.4 per gradient and 1.0 per second, a negative control.

Why not a full dense metric? Letting NUTS adapt a dense metric over every coordinate, without the rotation, is the direct alternative. In `Orthodont`, with $N = 27$ and 62 sampled coordinates, it recovered most of the gain per gradient (5.5–5.6 against 7.1 for the intercept and male) and did somewhat better on the variance components, but at twice the cost per run, so per second it gained 2.0–2.1 against the method’s 4.7. In the toenail model, with $N = 294$ and about 300 coordinates, it was worse than the method on every measure: a 300-dimensional metric estimated from 1000 warm-up iterations needed 47.6 leapfrog steps per draw against 21.2, per gradient it gained 3.2–3.7 against 5.5–6.7, and each run took 167 seconds against 7, so per second it was 0.2–0.3 times the *unrotated* fit. This is a result for the tested adaptation budget, not a limit of dense NUTS as such: a longer warm-up would estimate the metric better, at a cost that grows with the number of coordinates. The rotation is what makes a dense metric affordable: it moves the ridge into $p + kq$ coordinates (8 and 6 here), whatever N is.

Why not centre? In these two configurations the centred parameterization did not remove the degeneracy’s cost: per gradient it gave 0.4 on the `Orthodont` intercept and male coefficients and 0.8–1.1 on toenail’s. It also brought back the funnel that non-centring avoids. In `Orthodont`, with 4 observations per child, it produced 34 divergent transitions over the 3 seeds, $\hat{R}$ up to 1.047 and 52.5 leapfrog steps per draw, and the slope standard deviation mixed at 0.06 times the unrotated rate; in toenail, with binary outcomes, the random-effect standard deviation mixed at 0.35 times. Both carry little information per subject, the regime in which centring is known to do poorly (Gelfand et al., 1995; Browne et al., 2009); where subjects are informative, centring can work well, as JMBayes2’s centred coefficients do on the joint-model data of Section 8. The rotation needs no choice between the two parameterizations: it keeps the non-centred one and, in every design tested here, from these low-information settings to the information-dense designs of Section 6.4, reduced the location cost within it.

Across all arms, posterior means agreed with the unrotated fit’s to within 0.14 posterior standard deviations. Apart from the centred arm, $\hat{R} \leq 1.011$ and there were at most 2 divergences (full dense, `Orthodont`).

8 Real data

8.1 Rotated against unrotated

We fitted two standard joint-model datasets, both distributed with JMBayes2, twice from the same preliminary mixed-model and Cox fits and the same sampler seed, unrotated and rotated, with 4 chains of 1000 warm-up and 1000 sampling iterations and 3 seeds per dataset.

- **aids** (Abrams et al., 1994; Guo and Carlin, 2004): 467 patients, $\sqrt{\text{CD4}} \sim \text{obstime} + \text{drug} + \text{gender}$ with a random intercept and slope on `obstime`, and a hazard for death depending on drug. There are three absorbable intercept directions (constant, drug, gender) and one slope direction.

Table 5: Real data: ESS per second, rotated over unrotated, mean over 3 seeds (per-seed range in brackets), and sampler statistics.

	aids	pbc2
intercept	26.9 [21–34]	17.2 [12–25]
subject-constant covariates	24.8, 28.5 [21–33]	16.6 [12–23]
time slope	2.3 [2.0–2.7]	4.4 [4.0–4.6]
α	2.0 [1.8–2.4]	0.95 [0.88–1.01]
$\sigma_e, \sigma_b, \rho, \gamma$	1.4–2.4	0.82–1.28
sampling time (s), rotated / unrotated	10.8 / 17.6	12.4 / 11.7
leapfrog steps per iteration	31 / 92–111	60–63 / 63
maximum $\hat{R}$, all parameters	1.010 / 1.010	1.007 / 1.018

- **pbc2** (Murtaugh et al., 1994): 312 patients, $\log(\text{serum bilirubin}) \sim \text{year} + \text{age}$ with a random intercept and slope on year, and a hazard for death depending on drug. There are two absorbable intercept directions (constant, age) and one slope direction.

The intercept and subject-constant coefficients gained 17–28 times, and the time slope, also on an absorbable direction, 2.3–4.4 times (Table 5). Every posterior mean agreed between the two fits within Monte Carlo error (largest paired difference 1.65 Monte Carlo standard errors over 19 parameter-dataset pairs). The two datasets differ in one instructive way. On **aids** the unrotated trajectories were three times longer: the degeneracy was what set NUTS’s trajectory length, and removing it made every iteration cheaper, so every parameter gained, α included, and sampling time fell by 40%. On **pbc2** the trajectory length was set by something else, and the parameters the degeneracy does not involve changed by 0.82–1.28 times (α 0.95).

With a random intercept only, the same datasets show the degeneracy more starkly. Unrotated, β_0 and the subject-constant coefficients mixed at 0.04–0.07 ESS per draw, and 91–95% of their posterior variance was explained by the absorbable directions of the random effects; every other parameter mixed near one ESS per draw. The rotation raised the affected coefficients’ ESS per second 51–53 times on **aids** and 19–20 times on **pbc2** (2 seeds each), and the slowest parameter’s ESS per draw from 0.05 to 0.88 and from 0.06 to 0.78.

8.2 A blocked sampler

JMbayes2 (Rizopoulos et al., 2024) fits the same models by Metropolis-within-Gibbs and hierarchically centres the coefficients it can, which is the classical remedy for the location degeneracy (Gelfand et al., 1995). It therefore offers two comparisons: on the centred coefficients, of the rotation with centring; on the rest, and on α in particular, of HMC with a blocked sampler (Section 5). JMbayes2 was run with 4 chains of 16,000 iterations and 4,000 burn-in, well above its defaults, and its time is the whole model-fitting call. Each package’s ESS is its own native estimator, so the ratios compare two ESS estimators as well as two samplers, and we read them as a software-level comparison rather than a controlled benchmark of samplers.

Using each package’s native ESS, the unrotated NUTS fit trailed JMbayes2 by a factor of 17–

Table 6: ESS per second relative to JMBayes2 (same data, 3 seeds).

	aids		pbc2	
	unrotated	rotated	unrotated	rotated
intercept	0.04	0.97	0.05	0.89
subject-constant covariates	0.03–0.04	0.89–0.95	0.06	0.92
time slope	2.8	6.5	0.42	1.8
α	3.2	6.5	7.6	7.3
survival coefficient γ	1.1	2.7	2.1	1.8

33 on the coefficients JMBayes2 centres (Table 6). Rotated, it is at parity on them (0.89–0.97) and ahead on the others: 1.8–6.5 times on the time slope, 1.8–2.7 on γ and 6.5–7.3 on α . The rotation thus recovers what hierarchical centring buys on the regression coefficients, while keeping the non-centred parameterization for the random-effect variances. The lead on α is present without the rotation, and Section 5 attributes most of its per-draw part to blocked updating; the rotation added to it on **aids** (3.2 to 6.5) by shortening NUTS’s trajectories. JMBayes2 draws 4.7–7.1 times faster per second but keeps one effective draw in 29–36 for α (Figure 2).

9 Discussion

9.1 Related work

Approaches to a location degeneracy fall into three groups: remove the directions (constraints, projections, integration), reparameterize them (centring, non-centring, interweaving), or adapt the sampler’s metric to them. The rotation is a reparameterization chosen so that the third becomes cheap: in Section 7, a full dense metric without the rotation, adapted over the same 1000 warm-up iterations, was about 2 times (**Orthodont**) and about 20 times (**toenail**) less efficient per second on the intercept and subject-constant coefficients than the rotation with its small block, and centring did not remove the cost in those low-information settings. Unlike removal, it keeps the original parameter space and posterior and returns the original parameters.

Choosing a parameterization. That the parameterization of a hierarchical model decides how fast a sampler mixes is well established. Hierarchical centring (Gelfand et al., 1995, 1996) removes the location degeneracy by sampling the centred effects; the non-centred parameterization removes the funnel instead (Papaspiliopoulos et al., 2003, 2007); partial non-centring interpolates, and ASIS (Yu and Meng, 2011) interweaves the two within a Gibbs sampler. Gorinova et al. (2020) automate the choice per variable in probabilistic programs. All of these choose between centred and non-centred coordinates for each random effect. The rotation does not: it keeps the non-centred parameterization, with its benefits for the variances, and changes only which linear combinations of the standardized effects are sampling coordinates. In the comparison with JMBayes2, which centres the coefficients it can, this recovered what centring buys on those coefficients (Section 8). Browne

et al. (2009) studied three reparameterizations for multilevel survival models under Gibbs sampling. Their orthogonal parameterization orthogonalizes the fixed-effect predictors among themselves, as Stan’s QR reparameterization does (Stan Development Team, 2025), but they observe that predictors correlated with the cluster indicator vectors make fixed and random effects “play similar roles” and suggest orthogonality to those indicators as future work. The absorbable subspace is that span, handled by rotating the random effects rather than the predictors.

Removing the directions. A second family removes the absorbable directions. Sum-to-zero constraints are the classical example. Zanella and Roberts (2021) show that imposing such constraints always improves the Gibbs sampler for crossed random effects, and that it also helps NUTS dramatically on a Poisson crossed model (minimum ESS from 3.7 to 4977). Their construction samples the posterior conditioned on the constraint, and Stan’s `sum_to_zero_vector` likewise changes the prior unless compensated. Restricted spatial regression (Reich et al., 2006; Hughes and Haran, 2013) projects spatial random effects off the fixed-effect design, and is known to affect inference on the fixed effects (Hodges and Reich, 2010; Hanks et al., 2015; Khan and Calder, 2022). An earlier version of `jmlax` did the same for the absorbable directions: it swept them out of the sampled random-effect coordinates. Without a correction this miscalibrated the intervals for the swept coefficients; with an exact correction it was about as efficient as the rotation in ordinary designs and up to twice as efficient in the most information-dense ones (Section 6.4), at the price of the correction step. The closest precedent is Vines et al. (1996), whose transformation is exact: for independent Gaussian random effects and a Gaussian prior on the grand mean, the directions that shift the grand mean against the mean of each factor are a priori independent of everything else, appear in no likelihood term, and can be integrated out, leaving the grand mean plus the mean effects and sum-to-zero contrasts to sample. Their method covers the grand-mean direction only, reports contrasts rather than the original parameters, and leaves correlated random effects to further work; they also warn that the *ad hoc* alternative, recentring the effects inside each iteration, has an unknown effect on the stationary distribution. The rotation extends the exact approach to every absorbable direction, including the covariate-weighted ones and those of random slopes, to correlated random effects and to any prior on the fixed effects, and it returns the original parameters.

Analysing the directions. In the discussion of Zanella and Roberts (2021), Yang and Liu (2021) decompose the Gibbs sampler for a random-intercept model with covariates using the subject-level covariate means, which is the covariate-weighted absorbable subspace for the intercept column, and show that the convergence rate is governed by the correlation between β and the random effects along it. They use the decomposition to analyse Gibbs samplers, note that it breaks down for general designs, and do not treat random slopes; here the subspace is constructed exactly for every column from the design and used as sampling coordinates. In the same discussion Robert (2021) asked whether the effect of these parameterization choices had been studied for HMC, and the rejoinder (Zanella and Roberts, 2021) called that open. Lemma 1 and Proposition 4 are an analysis

on the HMC side for the absorbable directions: what a diagonal metric can and cannot do, what a dense block adds, and where a fixed metric falls short. They complement general results on linear preconditioning for MCMC (Hird and Livingstone, 2025): the block the rotation creates is small enough to precondition densely because the model says where the ridge is.

Joint-model software. JMbays2 (Rizopoulos et al., 2024) uses Metropolis-within-Gibbs; the function `stan_jm` in the `rstanarm` package (Brilleman et al., 2018) uses NUTS with non-centred random effects and so, we expect, meets the same ridge, which we have not tested; INLAjoint (Rustand et al., 2024) uses integrated nested Laplace approximations and does not sample. We have not found a previous treatment that applies the data-augmentation autocorrelation identity (Liu et al., 1994) to the association parameter together with a measurement of how its shared variance divides between the random effects and the baseline hazard.

9.2 Costs and limits

Two costs were measured and left as they are. With about 9 visits per subject, the rotation gave α 0.72–1.00 of the unrotated fit’s ESS per second, though it gained with denser follow-up and α remained far ahead of the blocked sampler either way; the mechanism is not established. And in the most information-dense stress designs, the swept variant was up to twice as efficient, as the fixed-metric residual of Section 4.4 predicts, although the rotation was still 5–30 times ahead of the unrotated fit there. The simulation evidence comes from one design family with a spline baseline hazard. The basis construction’s rank decision is a tolerance judgement when absorbable and identified directions are nearly collinear, although every returned direction is verified before use. Condition S is checked at each fit rather than assumed, and can fail for very flexible time bases with few observations per subject.

9.3 Extensions

A metric that follows the sampled σ_0 , for example by rescaling the dense block with the current value of L , is the obvious way to remove the residual of Section 4.4, and has not been tried. The construction needs only the fixed- and random-effects designs, so it applies to any model whose random effects enter through a linear predictor, including the GLMMs of `brms` and `rstanarm`, and could be offered there as a transformation of the non-centred random effects. Crossed random effects need one basis per grouping factor, each rotating that factor’s own index; the argument of Proposition 3 carries over unchanged.

9.4 Conclusion

The location degeneracy between fixed and random effects is shared by every regression with a random intercept and subject-constant fixed-effect columns, and it recurs for the time slope in every model with a random slope; this much is algebra. In the models studied here it slowed HMC

substantially on the coefficients an analyst reports. Rotating the standardized random effects so that the absorbable directions become a small block of coordinates, and giving that block and the fixed effects one dense metric block, largely whitens the resulting ridge without changing the model. In ESS per second the gains were 14 and 21 times for the intercept and time slope in simulation and 17–28 times for the intercept and subject-constant covariates on two joint-model datasets; in ESS per gradient evaluation they were 16 and 24 times in simulation and 5–7 times on the same coefficients in two ordinary mixed models. The association parameter of a joint model is a separate matter. In the two joint models measured, its posterior variance was largely shared with the survival parameters; a sampler that updates it conditionally pays for that sharing in autocorrelation, whereas HMC, heuristically, pays for it, if at all, in integration steps.

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A Proofs

Proof of Proposition 2. ($\supseteq$) Let $d \in \ker M_c$ and $v = v(d)$ as in Condition S. For $i \in S_c$, Proposition 1(b) gives $X_i d + v_i z_i = 0$; for $i \notin S_c$, $d \in \ker M_c$ gives $X_i d = 0$ and $z_i = 0$, so (L) holds. Condition S gives (T). Hence $v \in \mathcal{A}_c$.

($\subseteq$) Let $v \in \mathcal{A}_c$ with witness d . By (L) and Proposition 1(b), $d \in \ker M_c$ and $v_i = -z_i^\top X_i d / z_i^\top z_i$ for $i \in S_c$. For $i \notin S_c$, (T) at any t with $z_i(t) \neq 0$ gives $v_i = -x_i(t)^\top d / z_i(t)$, which by Condition S is $v_i(d)$. So $v = v(d)$.

If $z_i(t) = 0$ for every t as well as at every observation, then column c of subject i ’s random effects does not enter the likelihood at all, v_i is unrestricted, and the corresponding coordinate is identified by its prior alone; it plays no part in the ridge. $\square$

A sufficient condition for Condition S. Suppose each column of the fixed-effects design and column c of the random-effects design are, for subject i , functions $f_j(t; w_i)$ and $g(t; w_i)$ of time and baseline covariates, and that for each d and scalar s the function $\sum_j d_j f_j(\cdot; w_i) + s g(\cdot; w_i)$ lies in a linear space of functions that is determined by its values at subject i ’s n_i observation times (for example, polynomials of degree less than n_i). Then, for $i \in S_c$, (L) at the observation times implies the identity at all t , which is Condition S for that subject. Subject-constant columns, a linear time column and covariate-by-time interactions satisfy this with $n_i \geq 2$.

B Reproducibility

All code is in the `jmjax` repository. The development scripts below reproduce each result; paths are relative to the package root.

result	script (output)
Figure 1, Section 4.4 numbers	<code>dev/theory_rotation_toy.py</code> ; <code>dev/figures/make_figures.py</code>
Section 5, Gaussian check	<code>dev/theory_alpha_toy.py</code>
Section 5, R^2 and its split	<code>dev/alpha_missing_information.R</code>
Figure 2	<code>dev/figures/make_figures.py</code>
Table 1	<code>dev/pilot_rotate_grid.R</code> (<code>dev/pilot_rotate_grid.csv</code>)
Section 6.3	<code>ROT_GRID=q1 dev/pilot_rotate_grid.R</code> (<code>dev/pilot_rotate_q1.csv</code>)
Table 2	<code>ROT_GRID=stress dev/pilot_rotate_grid.R</code> (<code>dev/pilot_rotate_stress.csv</code>)
Section 6.3, runs to convergence	<code>dev/study_q1_longrun.R</code> (<code>dev/study_q1_longrun.csv</code>)
Figure 3	<code>dev/figures/make_figures.py</code>
Table 3	<code>dev/study_calibration.R</code>
α bias, $N = 1200$	<code>dev/study_alpha_bias.R</code> (<code>dev/alpha_bias_results.csv</code>)
Table 4	<code>dev/generalality/mixed_models.py</code> (<code>dev/generalality/results_orthodont_toenail_ext.csv</code>)
Tables 5 and 6	<code>dev/study_realdata_rotate.R</code>
Section 8, random intercept only	<code>dev/pilot_q1_intercept.R</code>
α against JMbates2, 1–12 covariates	<code>dev/study_orth_vs_jmbates2.R</code>
Example 1, basis checks	<code>dev/verify_absorbable_basis.py</code>